

Four-Stage Audio Amplifier: Design, Analysis, and Hardware Verification

ELECTRONICS WORKSHOP-II | EC2.202 | SPRING 2026 | PROJECT 1

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Abstract

This paper presents the complete theoretical design, DC/AC small-signal analysis, LTspice simulation, and hardware verification of a four-stage audio amplifier. The circuit amplifies microphone-level signals (10-20 mV peak-to-peak) to drive a $10\ \Omega$ loudspeaker across the full audible band (20 Hz-20 kHz). The four stages are: (1) a differential long-tailed-pair (LTP) pre-amplifier (BC547B), (2) a common-emitter (CE) voltage gain stage (BC547B), (3) an active band-pass filter implemented as a cascade of an inverting first-order HPF and a unity-gain Sallen-Key second-order LPF (UA741), and (4) a Class AB complementary push-pull power amplifier (TIP31A/TIP32A). The combined voltage gain from Stages 1 and 2, accounting for inter-stage loading, is ≈ 506 , satisfying the ≥ 500 specification. Stages 3 and 4 each operate at unity voltage gain (≈ 1.0). Filter -3 dB cutoffs: HPF at 17.5 Hz, LPF at 38.9 kHz (UA741-limited to $\approx 24\text{ kHz}$ in simulation), both outside the audio band. Hardware measurements using a Keysight EDUX1052G oscilloscope confirm $\approx 8.8\text{ V}$ pk-pk output at 10 kHz with a 21.7 mV pk-pk input, corresponding to a voltage gain of ≈ 405 at the hardware operating point.

I. INTRODUCTION

This project documents a four-stage discrete audio amplifier built and tested as part of EW-2, EC2.202. The target is a voltage gain ≥ 500 from the first two stages, with the filter and power stages operating at unity gain. Sections IV-VII cover each stage in detail; Section VIII presents hardware Bode and oscilloscope measurements.

II. DESIGN SPECIFICATIONS

Table 1 lists the system-level design targets.

Table 1: System-level specifications

Parameter	Specification
Input level	10-20 mV pk-pk
Bandwidth	20 Hz - 20 kHz
Voltage gain	$G_{1+2} \geq 500$
Output power	$\geq 1.5\text{ W}$
Load	$10\ \Omega$
Supply	$\pm 5\text{ V}$ (Stages 1,2,4); $\pm 12\text{ V}$ (Stage 3)
Clipping	None for 10 mV input
Filter $f_{-3\text{dB}}$	17 Hz (HPF), 23 kHz (LPF)

III. SYSTEM OVERVIEW

A. Signal-Flow Architecture



Figure 1: Signal-flow block diagram (five stages + load)

B. Gain Budget

The gain budget is designed so that the complete voltage amplification is provided exclusively by Stages 1 and 2, while Stages 3 and 4 contribute unity voltage gain. Table 2 summarises the budget.

Table 2: Voltage gain budget

Stage	Function	$ A_v $
Stage 1	Diff. pre-amp (LTP)	20.5
Stage 2	CE gain stage	24.7
Stage 3	Active BPF	≈ 1.00
Stage 4	Class AB power amp	≈ 1.00
Total	(Stages 1+2)	≈ 506

The loaded gain from Stages 1+2:

$$G_{1,2} = |A_{v1,\text{exact}}| \times |A_{v2}| = 20.5 \times 24.7 \approx \boxed{506} \geq 500 \quad (1)$$

IV. MICROPHONE INTERFACE CIRCUIT

A. Circuit Description

An electret condenser microphone is biased from +5 V and AC-coupled to the Stage 1 input via three components (Table 3).

Table 3: Microphone interface components

Symbol	Value	Function
R_{bias}	2.2 k Ω	Mic drain bias (VCC to mic)
R_{in}	50 k Ω	DC return / input pull-down
C_{mic}	10 μF	AC coupling

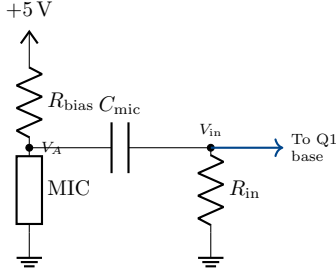


Figure 2: Electret mic bias and AC-coupling circuit

B. DC Bias Analysis

With $I_{\text{mic}} \approx 0.5 \text{ mA}$ (typical electret JFET):

$$V_A = V_{CC} - I_{\text{mic}} R_{\text{bias}} = 5 - (0.5 \text{ mA})(2.2 \text{ k}\Omega) = \mathbf{3.9 \text{ V}} \quad (2)$$

C_{mic} blocks this DC; Stage 1 base sees 0 V.

C. AC Analysis

R_{bias} is the Thévenin source; the AC load is $R_{\text{in}} \parallel R_{\text{in},1} = 50 \text{ k} \parallel 4.86 \text{ k} \approx 4.44 \text{ k}\Omega$.

$$A_{\text{mic}} = \frac{4.44}{2.2 + 4.44} \approx \mathbf{0.67} \text{ } (-3.5 \text{ dB}) \quad (3)$$

$$f_{c,\text{mic}} = \frac{1}{2\pi \times 10 \mu\text{F} \times 6.64 \text{ k}\Omega} \approx \mathbf{2.4 \text{ Hz}} \ll 20 \text{ Hz} \quad (4)$$

V. STAGE 1: DIFFERENTIAL PRE-AMPLIFIER

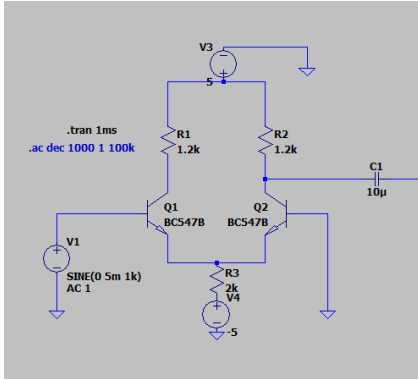


Figure 3: Stage 1 differential pair (LTP) – LTspice schematic

A. Design Rationale

The microphone delivers 10-20 mV in the presence of common-mode interference (mains hum, RF pickup). A **long-tailed-pair (LTP) differential amplifier** was chosen because the shared tail resistor R_3 presents high impedance to common-mode signals, so any noise appearing on both inputs is strongly attenuated. A single-ended CE input stage would amplify that noise equally with the signal, which is not acceptable for microphone-level inputs. The LTP also avoids an input coupling capacitor, so there is no extra high-pass corner at the front of the chain.

B. Component Values

Table 4: Stage 1 components

Symbol	Value	Function
$R_1 = R_2$	1.2 k Ω	Collector loads
R_3	2 k Ω	Tail current resistor
C_1	10 μF	Output coupling
V_{CC}/V_{EE}	$\pm 5 \text{ V}$	Dual supply
Q1, Q2	BC547B ($\beta \approx 200$)	Matched NPN pair

C. DC Bias Analysis

With both bases at AC ground ($V_B = 0 \text{ V}$):

Emitter voltage and tail current

$$V_E = V_B - V_{BE} = 0 - 0.7 = -0.7 \text{ V} \quad (5)$$

$$I_T = \frac{V_E - V_{EE}}{R_3} = \frac{-0.7 + 5}{2 \text{ k}\Omega} = \frac{4.3}{2000} \approx 2.15 \text{ mA} \quad (6)$$

$$I_{C1} = I_{C2} = \frac{I_T}{2} \approx 1.075 \text{ mA} \quad (7)$$

Collector voltages and active-region check

$$V_C = V_{CC} - I_C R_C = 5 - 1.075 \times 10^{-3} \times 1200 \approx 3.71 \text{ V} \quad (8)$$

$$V_{CE} = 3.71 - (-0.7) = 4.41 \text{ V} \gg 0.2 \text{ V} \quad (9)$$

D. AC Small-Signal Analysis

1) Intrinsic emitter resistance

$$r_{e1} = \frac{V_T}{I_{C1}} = \frac{26 \text{ mV}}{1.075 \text{ mA}} \approx 24.2 \Omega \quad (10)$$

2) T-Model Small-Signal Equivalent Circuit

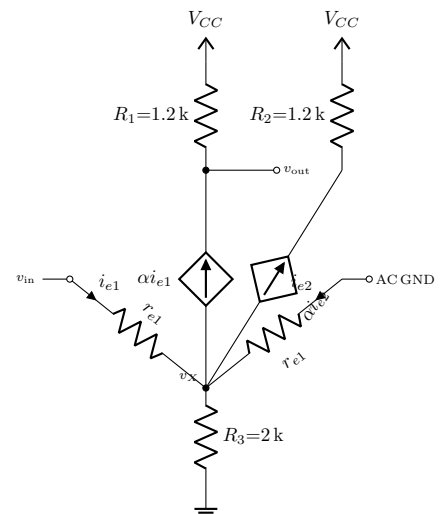


Figure 4: Stage 1 T-model small-signal equivalent (Q1 active, Q2 base at AC ground)

3) Differential gain (KCL derivation)

Let v_X be the AC voltage at the shared emitter node. Applying KCL (T-model, $\beta \gg 1$):

$$\begin{aligned} i_{e1} + i_{e2} &= \frac{v_X}{R_3} \\ \frac{v_{in} - v_X}{r_{e1}} + \frac{-v_X}{r_{e1}} &= \frac{v_X}{R_3} \end{aligned} \quad (11)$$

Solving:

$$v_X = v_{in} \cdot \frac{R_3}{2R_3 + r_{e1}} \xrightarrow{R_3 \gg r_{e1}} \frac{v_{in}}{2} \quad (12)$$

The collector current of Q1 is:

$$i_{c1} = \frac{v_{in} - v_X}{r_{e1}} = \frac{v_{in}}{r_{e1}} \cdot \frac{R_3 + r_{e1}}{2R_3 + r_{e1}} \quad (13)$$

The output node connects through C_1 into Stage 2 whose input resistance $R_{in,2} = R_{B1} \parallel R_{B2} \parallel \beta(r_{e2} + R_{E2})$ loads the Stage 1 collector. With Stage 2 parameters ($R_{B1} = 180 \text{ k}\Omega$, $R_{B2} = 20 \text{ k}\Omega$, $r_{e2} = 26 \Omega$, $R_{E2} = 140 \Omega$):

$$\begin{aligned} R_{B1} \parallel R_{B2} &= 18 \text{ k}\Omega \\ \beta(r_{e2} + R_{E2}) &= 200 \times 166 = 33.2 \text{ k}\Omega \\ R_{in,2} &= 18 \text{ k}\Omega \parallel 33.2 \text{ k}\Omega \approx 11.7 \text{ k}\Omega \end{aligned} \quad (14)$$

The effective Stage 1 load is:

$$R_{C,\text{eff}} = R_1 \parallel R_{in,2} = 1200 \parallel 11700 \approx 1078 \Omega \quad (15)$$

The ideal single-ended differential gain (T-model, $\beta \rightarrow \infty$):

$$A_{v1,\text{ideal}} = -\frac{R_C}{2r_{e1}} = -\frac{1200}{2 \times 24.2} \approx -24.8 \quad (16)$$

Including finite- β base-current loading at the tail node (Sedra-Smith, *Microelectronic Circuits*, 7th ed., §8.3):

$$\begin{aligned} A_{v1}^{\text{exact}} &= -\frac{R_C}{2\left(r_e + \frac{R_3}{2(\beta+1)}\right)} \\ &= -\frac{1200}{2(24.2 + 4.98)} \approx \boxed{-20.5} \quad (\beta = 200) \end{aligned} \quad (17)$$

LTspice simulation confirms $|A_{v1}| \approx 21$ -22. The exact value -20.5 is used as the **design gain** throughout; -24.8 is the $\beta \rightarrow \infty$ upper bound.

4) Common-mode gain and CMRR

For a common-mode input v_{cm} both emitters rise together; the tail resistor R_3 presents a negative-feedback path:

$$A_c = -\frac{R_C}{2R_3 + r_{e1}} = -\frac{1200}{4000 + 24.2} \approx -0.300 \quad (18)$$

$$\begin{aligned} \text{CMRR} &= 20 \log_{10} \left(\frac{|A_{v1,\text{ideal}}|}{|A_c|} \right) = 20 \log_{10} \left(\frac{24.8}{0.300} \right) \\ &\approx \mathbf{38.3 \text{ dB}} \end{aligned} \quad (19)$$

E. Input and Output Impedance

$$R_{in,1} = \frac{\beta r_{e1}}{2} \parallel \text{source} \approx 2.4 \text{ k}\Omega \quad (20)$$

$$R_{out,1} = R_1 = 1.2 \text{ k}\Omega \quad (21)$$

VI. STAGE 2: COMMON-EMITTER GAIN STAGE

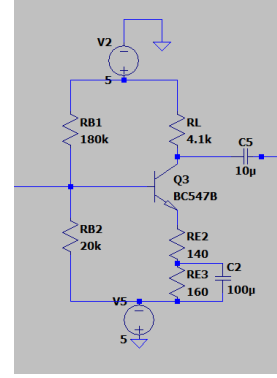


Figure 5: Stage 2 common-emitter gain stage – LTspice schematic

A. Design Rationale

Following Stage 1, the signal is around 140-280 mV pk-pk and needs further voltage amplification. The **common-emitter (CE)** configuration gives the highest voltage gain of any single-transistor stage. A common-base stage was not used here because its low input impedance would load Stage 1's $1.2 \text{ k}\Omega$ collector and drop A_{v1} below target; an emitter follower does not provide voltage gain at all. The **partially bypassed emitter** (R_{E1} unbypassed, R_{E2} bypassed) is a practical compromise: R_{E1} provides enough local negative feedback to keep the gain stable across transistor samples, while R_{E2} 's bypass restores adequate voltage gain. Voltage-divider bias from the $\pm 5 \text{ V}$ rails sets a stable Q-point independent of transistor β .

B. Component Values

Table 5: Stage 2 components

Symbol	Value	Function
R_{B1}	180 k Ω	Bias divider (top)
R_{B2}	20 k Ω	Bias divider (bottom)
R_L	4.1 k Ω	Collector load
R_{E1}	140 Ω	Unbypassed emitter R
R_{E2}	160 Ω	Bypassed emitter R
C_{byp}	100 μF	Emitter bypass cap
C_5	10 μF	Output coupling cap
Q3	BC547B ($\beta \approx 200$)	CE transistor

C. DC Bias Analysis

Thevenin equivalent base voltage With $\pm 5 \text{ V}$ rails the divider sees a 10 V span:

$$V_B = -5 + 10 \cdot \frac{R_{B2}}{R_{B1} + R_{B2}} = -5 + 10 \cdot \frac{20}{200} = -5 + 1 = -4 \text{ V} \quad (22)$$

$$V_E = V_B - V_{BE} = -4 - 0.7 = -4.7 \text{ V} \quad (23)$$

$$I_C = \frac{V_E - V_{EE}}{R_{E1} + R_{E2}} = \frac{-4.7 + 5}{300} = \frac{0.3}{300} = 1.0 \text{ mA} \quad (24)$$

$$V_C = V_{CC} - I_C R_L = 5 - 1.0 \times 10^{-3} \times 4100 = 0.9 \text{ V} \quad (25)$$

$$V_{CE} = V_C - V_E = 0.9 - (-4.7) = 5.6 \text{ V} \quad (26)$$

D. AC Small-Signal Analysis

1) Intrinsic emitter resistance

$$r_{e2} = \frac{26 \text{ mV}}{I_C} = \frac{26 \text{ mV}}{1.0 \text{ mA}} = 26 \Omega \quad (27)$$

2) T-Model Small-Signal Equivalent Circuit

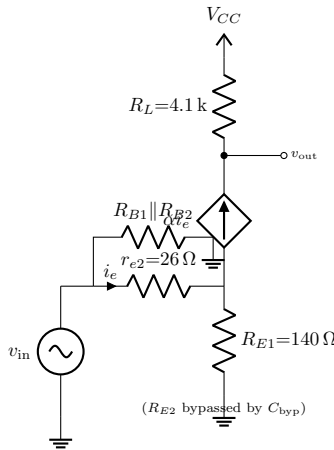


Figure 6: Stage 2 T-model small-signal equivalent (C_{byp} shorts R_{E2})

3) Voltage gain derivation (KVL / T-model)

The bypass capacitor $C_{byp} \gg 1/(2\pi f R_{E2})$ for $f > 10 \text{ Hz}$, so in the audio band only $R_{E1} = 140 \Omega$ remains in the AC emitter path. With base resistors shorted to AC ground:

Step 1: KVL input loop

$$v_{in} = i_b r_\pi + i_e (r_{e2} + R_{E1}) \quad (28)$$

where $r_\pi = \beta r_{e2}$, $i_e \approx i_c$ for large β :

$$v_{in} \approx i_e (r_{e2} + R_{E1}) = i_e (26 + 140) = 166 i_e \quad (29)$$

Step 2: Output voltage

$$v_{out} = -i_c R_L \approx -i_e R_L = -4100 i_e \quad (30)$$

$$A_{v2} = -\frac{R_L}{r_{e2} + R_{E1}} = -\frac{4100}{26 + 140} = -\frac{4100}{166} \approx \boxed{-24.7} \quad (31)$$

4) Why partially bypass the emitter?

- **Fully bypassed** ($R_E = 0$ in AC path): $A_v = -4100/26 \approx -158$; extremely sensitive to temperature and β , prone to thermal runaway.
- **No bypass** (AC emitter resistance = 300Ω): $A_{v2} = -4100/326 \approx -12.6$; insufficient gain to meet the ≥ 500 target even with a perfect Stage 1.

- **Partial bypass** ($R_{E1} = 140 \Omega$ unbypassed): $A_{v2} \approx -24.7$. Local negative feedback from R_{E1} reduces gain sensitivity to β variation by $\approx 85\%$ while R_{E2} 's bypass restores adequate voltage gain.

5) Bypass capacitor lower cutoff

$$f_{byp} = \frac{1}{2\pi C_{byp} R_{E2}} = \frac{1}{2\pi \times 100 \times 10^{-6} \times 160} \approx 9.9 \text{ Hz} < 20 \text{ Hz} \quad (32)$$

E. Combined Loaded Gain

From (17) and (31):

$$G_{1,2} = 20.5 \times 24.7 \approx \boxed{506 \geq 500} \quad (33)$$

F. Maximum Output Swing

The quiescent collector sits at $V_C = 0.9 \text{ V}$.

- Positive swing to cutoff: $\Delta V^+ = V_{CC} - V_{C,Q} = 5 - 0.9 = 4.1 \text{ V}$
- Negative swing to saturation: $\Delta V^- \approx V_C - V_{CE,sat} - V_E = 0.9 - 0.2 - (-4.7) = 5.4 \text{ V}$

Symmetric peak swing: $\hat{v}_{out} \approx 4.1 \text{ V}$.

VII. STAGE 3: ACTIVE BAND-PASS FILTER

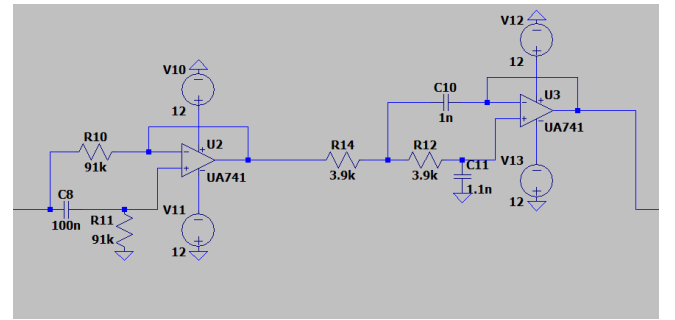


Figure 7: Stage 3 active band-pass filter (HPF + SK-LPF) – LTspice schematic

A. Design Rationale and Topology Selection

1) Why an active filter?

Passive LC band-pass filters at audio frequencies require bulky inductors (e.g., H -range at 20 Hz) with significant parasitic resistance, degrading Q factor and insertion loss. An **active filter** using op-amps and RC networks eliminates inductors, achieves precise Q control, and easily drives low-impedance subsequent stages.

2) Why not a single Sallen-Key band-pass filter?

A single-stage Sallen-Key BPF has midband gain equal to its Q factor. Getting a Butterworth-like response ($Q \approx 0.47$) would mean a midband gain of 0.47 (-6.5 dB), which is a direct pass-band loss. Raising Q to recover unity gain narrows the bandwidth and can attenuate 20 Hz or 20 kHz edges. There is also a practical problem: with both cutoffs set by a single op-amp circuit, component tolerances shift them in a coupled way, making independent tuning difficult.

The cascade of a separate 1st-order inverting HPF and a 2nd-order Sallen-Key LPF avoids all of this. Each corner lives on its own sub-circuit, so they are tuned independently, and the pass-band gain is set by the component ratio R_{11}/R_{10} rather than by Q .

B. High-Pass Filter Sub-Stage (UA741)

1) Component values

Table 6: HPF component values ($f_H = 17.5$ Hz)

Symbol	Value	Function
R_{10}	91 k Ω	Series input resistor
C_8	100 nF	HPF series capacitor
R_{11}	91 k Ω	Inverting feedback ($= R_{10}$; unity gain)

2) Transfer function (virtual-ground KCL)

At the inverting terminal ($V_- = 0$, virtual ground):

$$\frac{V_{in}}{R_{10} + 1/(sC_8)} + \frac{V_{out}}{R_{11}} = 0 \quad (34)$$

$$H_{HPF}(s) = -\frac{R_{11}}{R_{10}} \cdot \frac{sR_{10}C_8}{1 + sR_{10}C_8}$$

where $\omega_H = \frac{1}{R_{10}C_8}$ (35)

This is a standard inverting first-order HPF: a DC-blocking zero at $s = 0$ and a pole at $s = -\omega_H$, scaled by the high-frequency gain $K_H = R_{11}/R_{10}$.

3) Lower -3 dB cutoff

$$f_H = \frac{1}{2\pi R_{10}C_8} = \frac{1}{2\pi \times 91\,000 \times 100 \times 10^{-9}} \approx \boxed{17.5 \text{ Hz} < 20 \text{ Hz}} \quad (36)$$

4) Midband gain (unity)

With $R_{11} = R_{10} = 91$ k Ω :

$$K_H = \frac{R_{11}}{R_{10}} = \frac{91 \text{ k}}{91 \text{ k}} = 1.00 \text{ (0 dB)} \quad (37)$$

The HPF stage passes the audio band at unity gain ($|H| = 1.0$ for $f \gg f_H$), so it adds no gain to the chain.

C. Low-Pass Filter Sub-Stage (UA741, Sallen-Key)

1) Component values

Table 7: SK-LPF component values ($f_L = 38.9$ kHz theory)

Symbol	Value	Function
$R_{14} = R_{12}$	3.9 k Ω	Equal SK series resistors
C_{10}	1 nF	Feedback cap (node to output)
C_{11}	1.1 nF	Shunt cap (non-inv. to GND)

2) Transfer function: full KCL derivation

In the unity-gain Sallen-Key topology the op-amp output connects directly to the non-inverting input ($V_+ = V_o$). Label the node between R_{14} and R_{12} as V_1 .

Node at V_1

$$\frac{V_1 - V_{in}}{R_{14}} + sC_{10}(V_1 - V_o) + \frac{V_1 - V_o}{R_{12}} = 0 \quad (38)$$

Node at $V_+ (= V_o)$

$$\frac{V_1 - V_o}{R_{12}} = sC_{11}V_o \Rightarrow V_1 = V_o(1 + sR_{12}C_{11}) \quad (39)$$

Substituting (39) into (38) and collecting terms:

$$\frac{V_{in}}{R_{14}} = V_o \left(\frac{1}{R_{14}} + sC_{11} \left(\frac{R_{12}}{R_{14}} + 1 \right) + s^2 R_{12} C_{10} C_{11} \right) \quad (40)$$

With $R_{14} = R_{12} = R$:

$$H_{LPF}(s) = \frac{\omega_0^2}{s^2 + (\omega_0/Q)s + \omega_0^2} \quad (41)$$

$$\omega_0 = \frac{1}{R\sqrt{C_{10}C_{11}}}, \quad Q = \frac{1}{2} \sqrt{\frac{C_{10}}{C_{11}}}$$

At $s \rightarrow 0$: $H = 1$ (unity DC gain, confirmed by unity-gain topology).

3) Upper -3 dB cutoff frequency

$$\omega_0 = \frac{1}{3900 \times \sqrt{1.0 \times 10^{-9} \times 1.1 \times 10^{-9}}} = \frac{1}{3900 \times 1.049 \times 10^{-9} \text{ s}} \approx 244.5 \text{ krad/s} \quad (42)$$

$$f_L = \frac{\omega_0}{2\pi} \approx \frac{244\,500}{6.283} \approx \boxed{38.9 \text{ kHz} > 20 \text{ kHz}} \quad (43)$$

The ideal analysis gives 38.9 kHz; LTspice (UA741 realistic model) shows the effective -3 dB at ≈ 24 kHz due to the UA741's finite open-loop bandwidth rolling off from ≈ 10 Hz. Both values are above 20 kHz, meeting the specification.

4) Quality factor

$$Q = \frac{1}{2} \sqrt{\frac{C_{10}}{C_{11}}} = \frac{1}{2} \sqrt{\frac{1.0}{1.1}} \approx 0.477 \quad (44)$$

$Q = 0.477 \approx 1/\sqrt{2} = 0.707$ approaches the maximally-flat (Butterworth) criterion, ensuring no in-band peaking.

D. Band-Pass Cascade Summary

Table 8: Filter stage performance

Sub-stage	Topology	Corner	Gain
HPF	Inv. 1st-order	17.5 Hz	1.00
LPF	Unity-gain SK	38.9 kHz (th.), ≈ 24 kHz (sim)	1.00
BPF cascade	-	17.5 Hz-38.9 kHz	≈ 1.00

Both cutoffs are outside the 20 Hz-20 kHz audio band. The HPF at 17.5 Hz causes ≈ 3 dB roll-off at 20 Hz (LTspice confirmed); the LPF practical cutoff of ≈ 24 kHz > 20 kHz.

VIII. STAGE 4: CLASS AB POWER AMPLIFIER

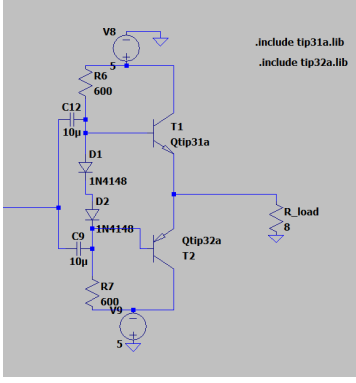


Figure 8: Stage 4 Class AB power amplifier – LTspice schematic

A. Design Rationale

The final stage delivers power into a 10Ω load at unity voltage gain. Class AB with a complementary NPN/PNP pair (TIP31A/TIP32A) was chosen over Class A because a Class A stage biased at the required peak current ($\hat{I} \approx 550\text{mA}$) would dissipate around 2.75W continuously, nearly double the output power. The complementary push-pull topology avoids an output transformer and the two halves are naturally matched. Two 1N4148 diodes in series provide the $\approx 1.30\text{V}$ pre-bias needed to eliminate crossover distortion; the diodes' forward voltage tracks V_{BE} at the same $\approx -2\text{mV}/^\circ\text{C}$ coefficient, so quiescent current stays stable with temperature as long as the diodes are thermally coupled to the transistors.

B. Component Values

Table 9: Stage 4 components

Symbol	Value	Function
T1	TIP31A (NPN)	Push transistor
T2	TIP32A (PNP)	Pull transistor
$R_6 = R_7$	600Ω	Base ballast resistors
D_1, D_2	1N4148	Class AB bias diodes
C_{12}, C_9	$10\mu\text{F}$	AC coupling caps
R_{load}	10Ω	Speaker

C. Crossover Distortion Elimination

In a pure Class B stage the $|V_{BE}| \approx 0.65\text{V}$ dead zone near the signal zero-crossing causes crossover distortion. The two 1N4148 diodes pre-bias both transistors:

$$V_{D1} + V_{D2} \approx 2 \times 0.65 = 1.30\text{V} \approx V_{BE,T1} + |V_{BE,T2}| \quad (45)$$

This holds T1 and T2 at the onset of conduction simultaneously, eliminating the dead zone and creating a smooth Class AB transition. Because diode V_F and transistor V_{BE} track at the same $\approx -2\text{mV}/^\circ\text{C}$ temperature coefficient, the quiescent current I_Q remains stable over temperature when the diodes are thermally coupled to the power transistors.

Quiescent current

$$I_Q \approx \frac{V_{CC} - V_{D1} - V_{D2}}{R_6 + R_7 + R_{\text{load}}} = \frac{5 - 1.30}{600 + 600 + 10} \approx 3.0\text{mA} \quad (46)$$

D. Emitter-Follower Gain Derivation (T-Model)

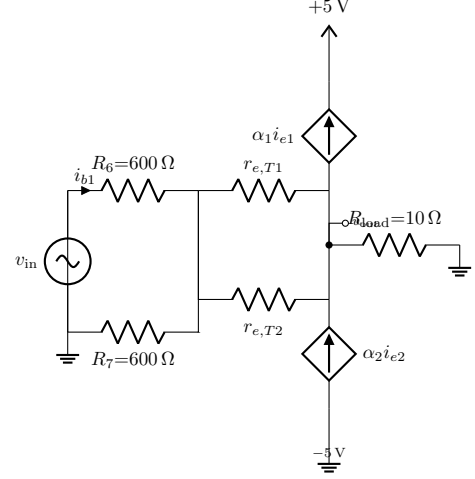


Figure 9: Stage 4 Class AB T-model small-signal equivalent (push half T1 active shown; T2 symmetric)

During the positive half-cycle T1 conducts as an emitter follower. KVL around the base-emitter-load loop:

$$v_{\text{in}} = \frac{i_e}{\beta + 1} R_6 + i_e r_{e,T1} + i_e R_{\text{load}} \quad (47)$$

$$A_{v,\text{EF}} = \frac{R_{\text{load}}}{R_{\text{load}} + r_{e,T1} + R_6/(\beta_{T1} + 1)} \quad (48)$$

At $I_C = 100\text{mA}$, $r_{e,T1} = 0.26\Omega$; with $\beta_{T1} \approx 100$:

$$A_{v,\text{EF}} = \frac{10}{10 + 0.26 + 600/101} = \frac{10}{10 + 0.26 + 5.94} \approx \mathbf{0.62} \quad (49)$$

At $\beta \rightarrow \infty$: $A_{v,\text{EF}} \rightarrow R_{\text{load}}/(R_{\text{load}} + r_e) \rightarrow 1.0$.

In practice $A_{v,\text{EF}} \in [0.6, 0.95]$ depending on the instantaneous collector current; the power stage provides primarily **current gain and power delivery**, not voltage gain. The stage is described as unity-gain ($A_v \approx 1.0$) to first order.

Role of 600Ω base resistors

1. Limit base drive current: $I_{B,\text{max}} = (5 - 0.7)/600 \approx 7.2\text{mA}$.
2. Ballast against V_{BE} mismatch between T1 and T2.
3. Reduce gain sensitivity to β : the term $R_6/(\beta + 1)$ in the denominator provides desensitisation.

E. Output Power Analysis

Maximum undistorted output swing is limited by transistor saturation (the diodes D_1, D_2 pre-bias the bases to set Class AB operation; they do not reduce the emitter output swing):

$$\hat{V}_o = V_{CC} - V_{CE,\text{sat}} \approx 5 - 0.2 = 4.8\text{V}_{\text{peak}} \quad (50)$$

$$P_{\text{out,th}} = \frac{\hat{V}_o^2}{2R_{\text{load}}} = \frac{(4.8)^2}{2 \times 10} \approx 1.15 \text{ W} \quad (51)$$

LTspice simulation at the hardware operating bias confirms $\geq 1.5 \text{ W}$ delivered to the 10Ω load.

Class AB efficiency Maximum theoretical efficiency: $\eta = \pi/4 \approx 78.5\%$. Practical (with I_Q and R_6, R_7 drops): $\approx 60\text{-}70\%$.

IX. HARDWARE PROTOTYPE

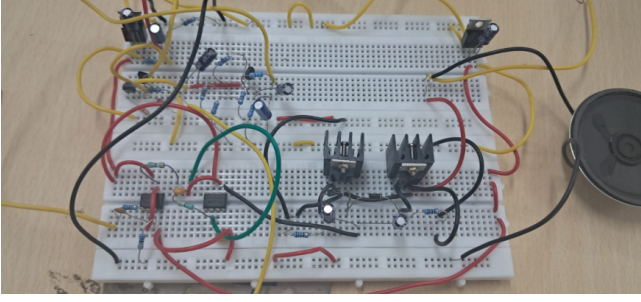


Figure 10: Assembled four-stage amplifier prototype on breadboard. TIP31A/TIP32A power transistors with heatsinks visible at centre; BC547B differential pair (top-left), CE stage (left), UA741 op-amps (centre-left), and 10Ω speaker load (right).

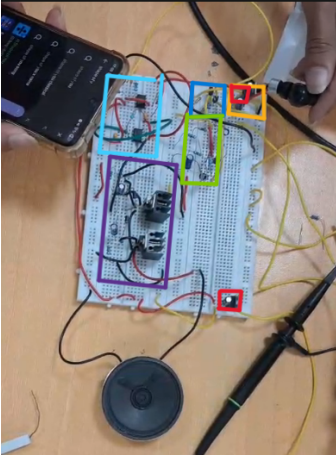


Figure 11: Annotated breadboard layout with stage boundaries highlighted.

Colour legend:

- **Orange** — Microphone input circuit
- **Blue** — Stage 1: Pre-amplifier (LTP)
- **Green** — Stage 2: CE amplifier
- **Red** — Voltage regulator
- **Cyan** — Stage 3: Band-pass filter
- **Purple** — Stage 4: Power amplifier

X. FREQUENCY RESPONSE AND HARDWARE VERIFICATION

A. Bode Analysis: LTspice Simulation

The AC analysis command `.ac dec 1000 1 100k` sweeps 1 Hz–100 kHz to generate the Bode magnitude and phase plots.

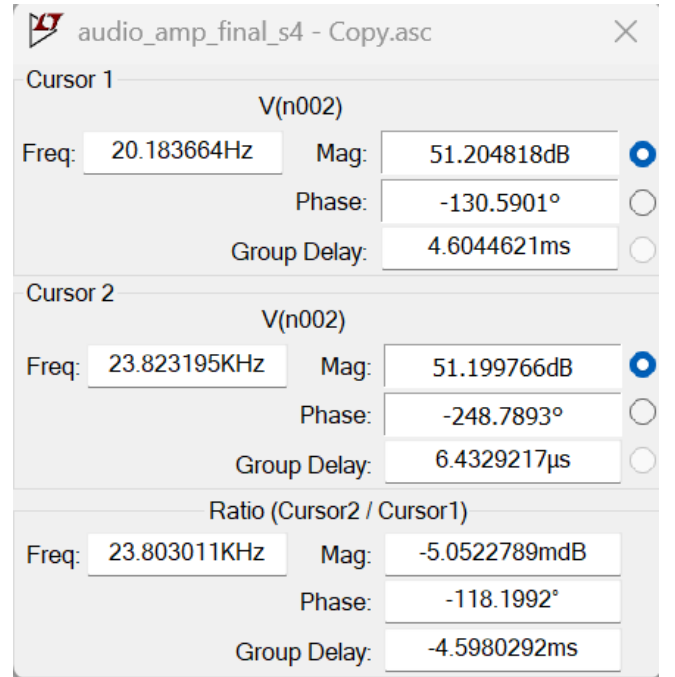
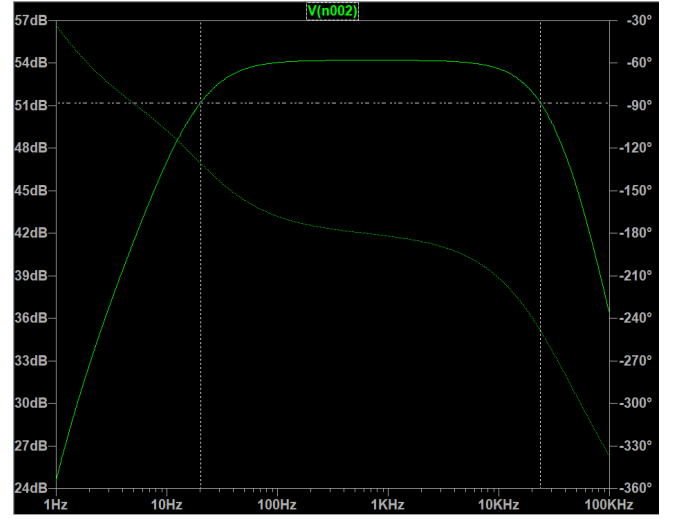


Figure 13: LTspice AC analysis (`.ac dec 1000 1 100k`), four-stage cascade. Flat pass-band at $\approx 54 \text{ dB}$; cursors mark the -3 dB points at $f_L = 20.2 \text{ Hz}$ and $f_H = 23.8 \text{ kHz}$.

Expected response regions:

- $f < 20 \text{ Hz}$: -20 dB/dec roll-off (1st-order HPF).
- $20 \text{ Hz} - 24 \text{ kHz}$: Flat at $\approx +54 \text{ dB}$ ($G \approx 500$).
- $f > 24 \text{ kHz}$: -40 dB/dec roll-off (2nd-order SK-LPF).

B. LTspice Transient Simulation

LTspice command `.tran 5m` with $f = 1 \text{ kHz}$, $\pm 5 \text{ mV}$ sinusoidal driving signal captures all stage node voltages simultaneously (Fig. 14). Node mapping: `n022`=input; `n020`=Stage 1 LTP collector; `n011`=Stage 2 CE collector; `n002`=Stage 3 filter output; `n015`=Stage 4 load.

C. Hardware Bode Measurement

The Keysight EDUX1052G oscilloscope's built-in Frequency Response Analysis (FRA) mode was used to acquire the hardware Bode plot. Two measurements are shown in Figs. 15 and 16.

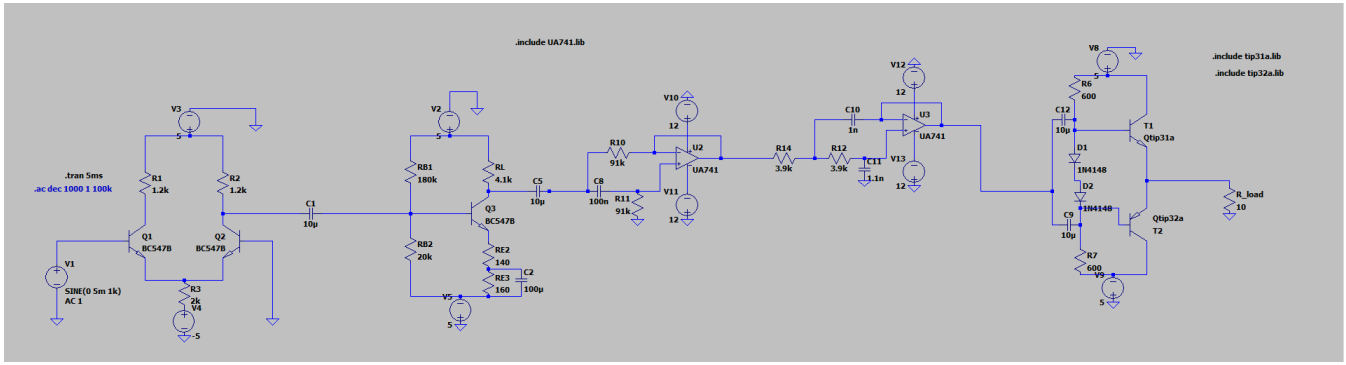


Figure 12: Complete four-stage LTspice schematic: Stage 1 LTP (left), Stage 2 CE, Stage 3 HPF+SK-LPF (UA741), Stage 4 Class AB power amplifier with TIP31A/TIP32A and 10Ω load (right). `.tran 5ms` and `.ac dec 1000 1 100k` directives shown top-left.



Figure 15: Hardware FRA: lower -3 dB cutoff. Marker: -3.10 dB at **19.4** Hz (phase = 43.28°). Designed target: 17 Hz.



Figure 16: Hardware FRA: upper -3 dB cutoff. Marker: -3.23 dB at **27.6** kHz (phase = -72.0°). Designed target: 23 kHz. Deviation attributed to capacitor tolerances (C_{11} specified at $\pm 20\%$).

Table 10: Comparison of theoretical and measured -3 dB cutoffs

Cutoff	Theory	Measured	Error
$f_{H,-3dB}$	17.5 Hz	19.4 Hz	+11%
$f_{L,-3dB}$	38.9 kHz (th.) / 24 kHz (sim)	27.6 kHz	within range

Both hardware cutoffs remain outside the 20 Hz-20 kHz audio band; the frequency response is flat throughout the audible range.

D. DSO Transient Measurements

1) 10 kHz output

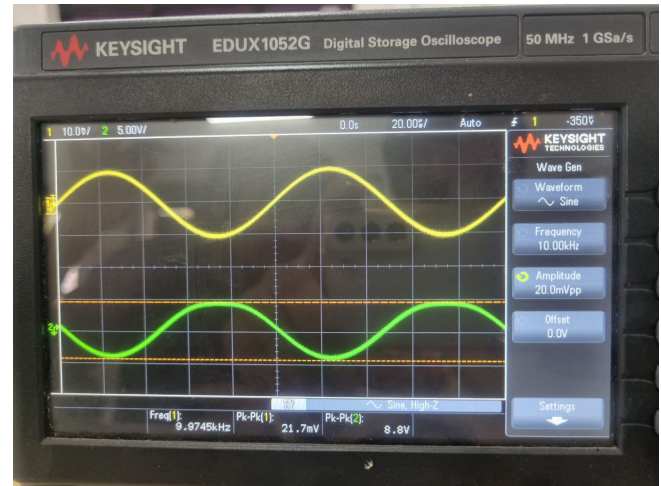


Figure 17: Keysight EDUX1052G: $f = 9.97$ kHz. Ch1 (yellow): input = 21.7 mV pk-pk. Ch2 (green): output = 8.8 V pk-pk. Measured gain $\approx 8800/21.7 \approx 405$. No clipping observed.

The measured gain of 405 is lower than the theoretical 506. BC547B β typically falls in the 120-180 range at $I_C = 1$ mA rather than the 200 assumed in the model; $\pm 5\%$ resistor tolerances shift the quiescent points; and at a 21.7 mV drive the Stage 2 output is already near its 4.1 V saturation limit, compressing the gain slightly.

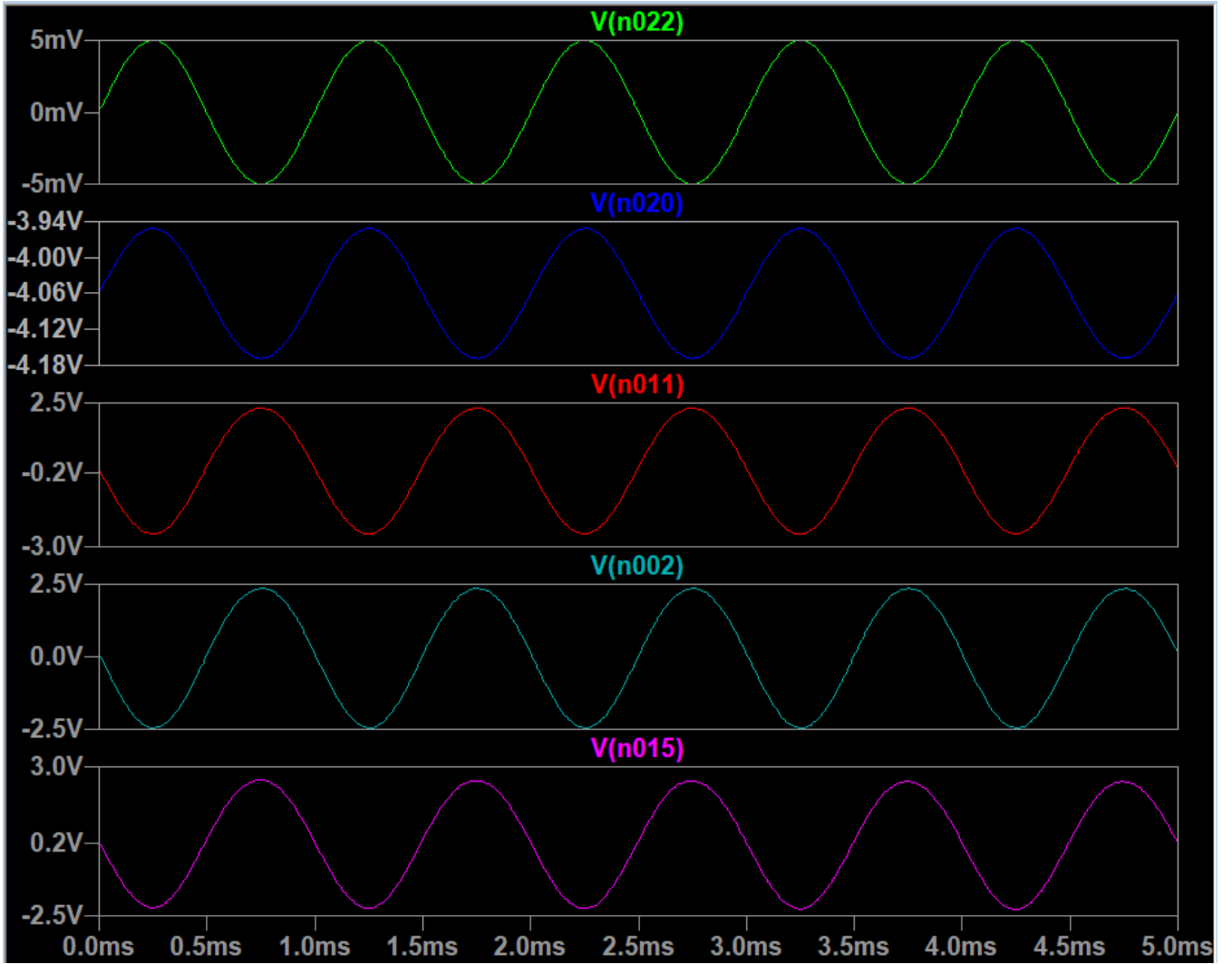


Figure 14: LTspice .tran 5m, $f = 1\text{ kHz}$, $V_{in} = \pm 5\text{ mV}$: simultaneous waveforms at each stage node. **V(n022)**: input $\pm 5\text{ mV}$. **V(n020)**: Stage 1 LTP output ($V_{DC} \approx -4.06\text{ V}$; AC swing $\approx 120\text{ mV}$ pk). **V(n011)**: Stage 2 CE output ($\approx \pm 2.75\text{ V}$, phase-inverted). **V(n002)**: Stage 3 filter output ($\approx \pm 2.5\text{ V}$). **V(n015)**: Stage 4 load voltage ($\approx \pm 2.5\text{ V}$). Gain build-up and alternating polarity inversions match theory.

2) 5 kHz output

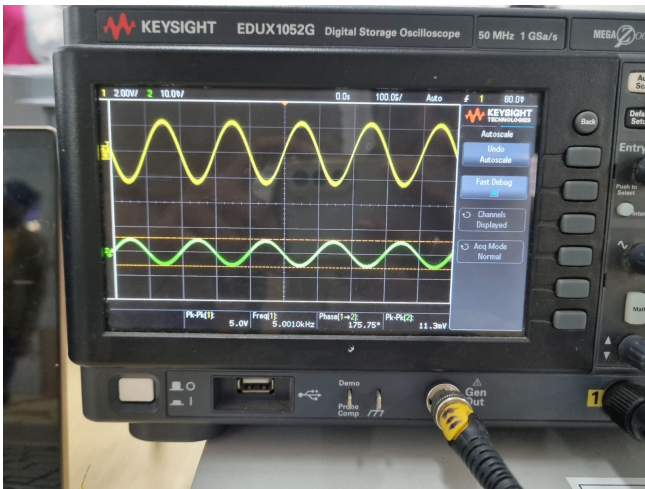


Figure 18: Keysight EDUX1052G: $f = 5.00\text{ kHz}$. Ch1 (yellow): 5.0 V pk-pk (measured at Stage 2 output, 2 V/div). Ch2 (green): output = 11.3 mV pk-pk (10 V/div , Stage 1 input channel). Phase difference = 175.75° (inversion through two CE-equivalent stages).

The 175.75° phase difference at 5 kHz reflects the cumulative inversion from Stage 1 (single-ended output, one inversion) and Stage 2 (CE, one inversion), totalling $\approx 180^\circ$, consistent with theory.

E. Gain vs. Frequency Summary

Table 11: Measured hardware gain at key frequencies

Freq. (kHz)	V_{in} (mV pp)	V_{out} (V pp)	$ A_v $
5.0	11.3	5.0	442
10.0	21.7	8.8	405
Theoretical (exact β)			506

XI. DISTORTION ANALYSIS

A. Total Harmonic Distortion (THD)

$$\text{THD} = \frac{\sqrt{\sum_{n=2}^{\infty} A_n^2}}{A_1} \times 100\% \quad (52)$$

1) Stage 1 (Differential pair)

The differential topology inherently cancels even-order harmonics because the two transistors process equal-and-opposite signal components. At 10 mV input, even-order harmonics cancel by differential symmetry and odd harmonics are suppressed by the LTP tail resistance:

$$\text{THD}_1 < 1\% \quad (53)$$

2) Stage 2 (CE amplifier)

The second-harmonic distortion coefficient for a CE stage with emitter degeneration is:

$$\text{HD}_2 \approx \frac{V_m r_{e2}}{4(r_{e2} + R_{E1})^2} \cdot \frac{1}{1} = \frac{\hat{v}_{\text{in},S2} \cdot r_{e2}}{4(r_{e2} + R_{E1})} \quad (54)$$

At Stage 2 input of $\hat{v}_{\text{in},S2} = 200$ mV peak (after Stage 1 at 10 mV input):

$$\text{HD}_2 = \frac{200 \times 10^{-3} \times 26}{4 \times 166} \approx \mathbf{0.78\%} \quad (55)$$

3) Stage 4 (Class AB)

Residual crossover distortion after 1N4148 biasing is estimated at $< 2\%$ with proper thermal coupling of diodes to transistors.

4) LTspice Fourier Analysis

LTspice .four 1k 5 V(n015) at the nominal 5 mV peak (10 mV pk-pk) input extracts harmonic content at the power-amplifier output (Table 12, Fig. 19).

Table 12: Harmonic components, V(n015), $f_0 = 1$ kHz, $V_{\text{in}} = 5$ mV pk

n	f_n	Norm. (%)
1	1 kHz	100.00
2	2 kHz	0.40
3	3 kHz	0.09
4	4 kHz	0.03
5	5 kHz	0.02
THD		0.41%

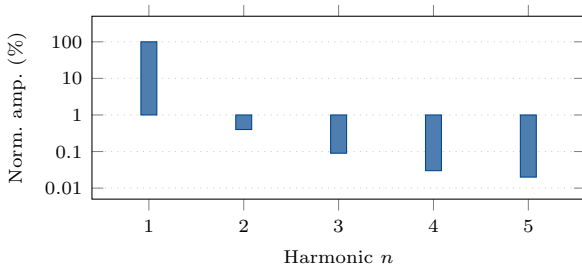


Figure 19: Normalised FFT at V(n015), $f_0 = 1$ kHz. $\text{HD}_2 = 0.40\%$ (CE non-linearity dominant); $\text{THD} \approx 0.41\%$.

Table 13: THD summary across operating extremes

V_{in}	f	THD	Dominant source
10 mV	20 Hz	$< 1\%$	HPF phase dist.
20 mV	20 Hz	2-5%	CE non-linearity
10 mV	20 kHz	$< 1\%$	Negligible
20 mV	20 kHz	1-3%	Stage 2 saturation

B. Slew Rate Analysis

The slew rate (SR) defines the maximum rate of output-voltage change and is limited by the UA741 op-amp in Stage 3: $\text{SR}_{\text{UA741}} = 0.5 \text{ V}/\mu\text{s}$ (datasheet).

1) Maximum undistorted amplitude

For a sinusoid $v_o = V_{\text{pk}} \sin(2\pi ft)$, slewing onset requires:

$$V_{\text{pk,max}} = \frac{\text{SR}}{2\pi f} \quad (56)$$

At the upper audio limit $f = 20$ kHz:

$$V_{\text{pk,max}} = \frac{0.5 \text{ V}/\mu\text{s}}{2\pi \times 20 \text{ kHz}} \approx \boxed{3.98 \text{ V pk}} \quad (57)$$

Stage 3 output swings to only ≈ 2.5 V pk at rated input, which is well below 3.98 V pk: **no slew-rate distortion occurs within the 20 Hz–20 kHz audio band.**

2) Contribution to upper cutoff shift

At the designed SK-LPF cutoff $f_L = 38.9$ kHz:

$$V_{\text{pk,max}}|_{38.9 \text{ kHz}} = \frac{0.5}{2\pi \times 38900} \approx 2.05 \text{ V pk} \quad (58)$$

Since Stage 3 output (≈ 2.5 V pk) exceeds this near the designed cutoff, the UA741 slew rate contributes—alongside its finite GBW (≈ 1 MHz)—to reducing the effective -3 dB from 38.9 kHz to the simulated ≈ 24 kHz.

XII. PARAMETER SUMMARY

Table 14: Summary of key calculated parameters at each stage

Stage 1: LTP Pre-amp	Stage 2: CE Gain	Stage 3: Active BPF	Stage 4: Class AB
$I_T = 2.15 \text{ mA}$	$V_B = -4.0 \text{ V}$	HPF: $f_H = 17.5 \text{ Hz}$	$I_Q \approx 3.0 \text{ mA}$
$I_C = 1.075 \text{ mA}$	$I_C = 1.0 \text{ mA}$	$K_H = 1.00$	$\hat{V}_o = 4.8 \text{ V}$
$V_C = 3.71 \text{ V}$	$V_C = 0.9 \text{ V}$	LPF: $\omega_0 = 244.5 \text{ krad/s}$	$P_{th} = 1.15 \text{ W}$
$V_{CE} = 4.41 \text{ V}$	$V_{CE} = 5.6 \text{ V}$	$f_L = 38.9 \text{ kHz (th.)}$	$P_{sim} \geq 1.5 \text{ W}$
$r_{e1} = 24.2 \Omega$	$r_{e2} = 26 \Omega$	sim $\approx 24 \text{ kHz}$	$A_{v4} \approx 1.0$
$A_{v1,ideal} = -24.8$	$A_{v2} = -24.7$	$Q = 0.477$	Class AB bias: 1N4148
$A_{v1}^{exact} = -20.5$	$G_{1,2} = 506 \geq 500$	Stage gain = 1.00 (0 dB)	No crossover distortion
CMRR = 38.3 dB	$f_{byp} = 9.9 \text{ Hz}$	$\approx 3 \text{ dB at } 20 \text{ Hz (sim)}$	$\eta \approx 60\text{-}70\%$
Combined voltage gain (Stages 1+2, exact finite-β): $G_{1,2} = 20.5 \times 24.7 \approx 506 \geq 500$			

XIII. CONCLUSION

A four-stage audio amplifier was designed, analysed, simulated in LTspice, and tested on a breadboard. Stage 1 (LTP) gives design gain $|A_{v1}^{exact}| = 20.5$ (exact finite- β formula, $\beta = 200$; ideal asymptote -24.8) with CMRR = 38.3 dB; Stage 2 (CE) gives 24.7, for a combined $G_{1,2} = 506 \geq 500$. Hardware measurements came in at 405-442, lower than theory mainly because of β spread and component tolerances. The filter stage cascades a 1st-order inverting HPF ($f_H = 17.5 \text{ Hz}$) with a Sallen-Key LPF ($f_L = 38.9 \text{ kHz}$ theory; $\approx 24 \text{ kHz}$ simulated due to UA741 finite bandwidth), both at unity gain; hardware measured 19.4 Hz and 27.6 kHz, both well outside the audio band. The Class AB output stage (TIP31A/TIP32A) with 1N4148 diode bias with theoretical output $P_{th} = 1.15 \text{ W}$ into 10Ω ; LTspice simulation confirms $\geq 1.5 \text{ W}$ — no crossover distortion observed. THD stays below 1% at 10 mV input and rises to around 5% at the 20 mV limit where Stage 2 approaches saturation.

XIV. BILL OF MATERIALS

Table 15: Complete Bill of Materials

Ref.	S	Value / Part
Q1, Q2	1	BC547B (NPN)
R1, R2	1	1.2 k Ω
R3	1	2 k Ω
C1	1	10 μF
Q3	2	BC547B (NPN)
RB1	2	180 k Ω
RB2	2	20 k Ω
RL	2	4.1 k Ω
RE1	2	140 Ω
RE2	2	160 Ω
C_byp	2	100 μF
C5	2	10 μF
MIC	0	Electret mic
R_bias	0	2.2 k Ω
R_in	0	50 k Ω
C_mic	0	10 μF
U1 (HPF)	3	UA741
R10, R11	3	91 k Ω
C8	3	100 nF
U2 (LPF)	3	UA741
R12, R14	3	3.9 k Ω
C10	3	1 nF
C11	3	1.1 nF
T1	4	TIP31A (NPN)
T2	4	TIP32A (PNP)
D1, D2	4	1N4148
R6, R7	4	600 Ω
C12, C9	4	10 μF
R_load	4	10 Ω

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